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S2IGAN: Speech-to-Image Generation via Adversarial Learning

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https://github.com/xinshengwang/S2IGAN

https://xinshengwang.github.io/project/s2igan/

Background

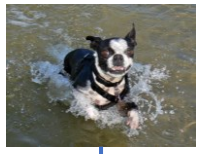
Cross-modal learning between visual and speech

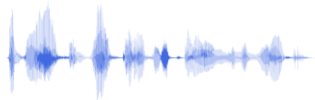
Identify matching pairs

"A golden brown dog is jumping over a barrier."

"A black and white dog is playing in the water. "

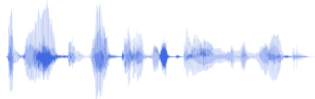
Describe image using speech



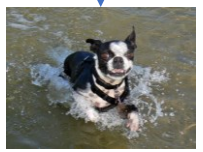


"A black and white dog is playing in the water. "

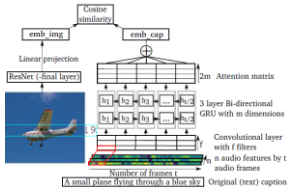
Imagine the image according to the speech description



"A black and white dog is playing in the water. "

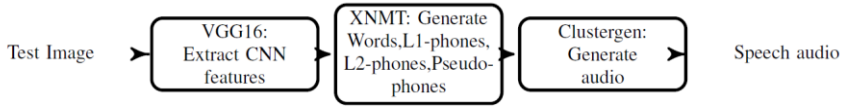


Cross-modal retrieval



Interspeech 2019, Merkx et al.

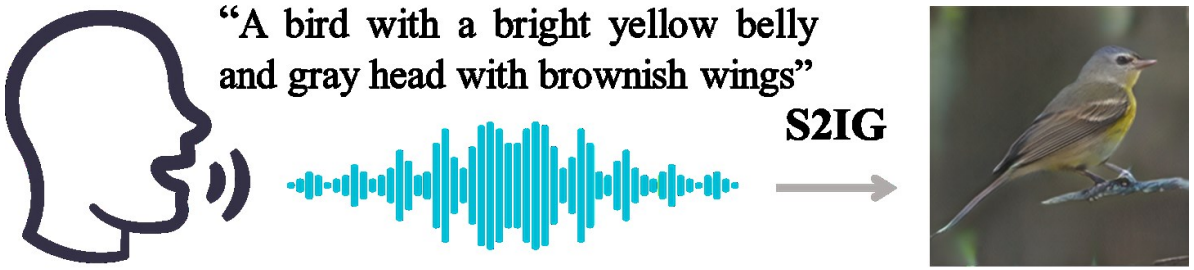
Image-to-speech generation



ICNLSSP 2017, Hasegawa-Johnson et al.

New task

Speech-to-image generation (S2IG)

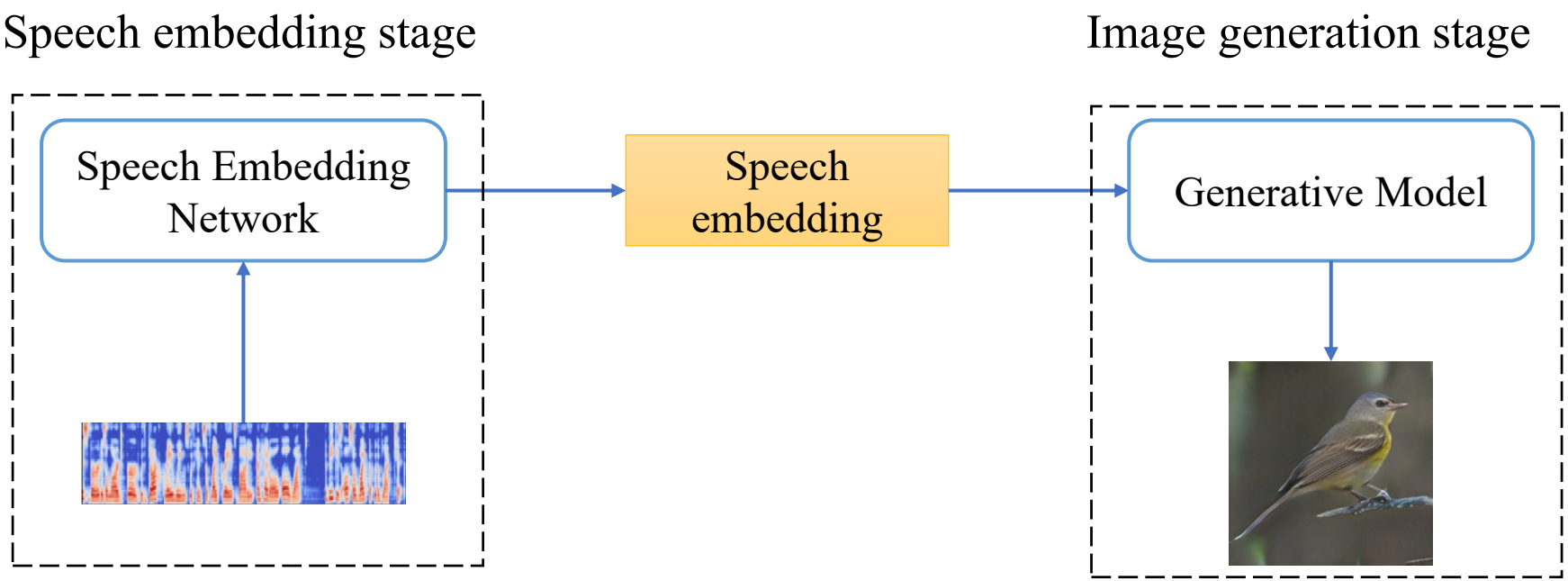


S2IG: generate images on the basis of spoken descriptions bypassing any text information

Method

Overview

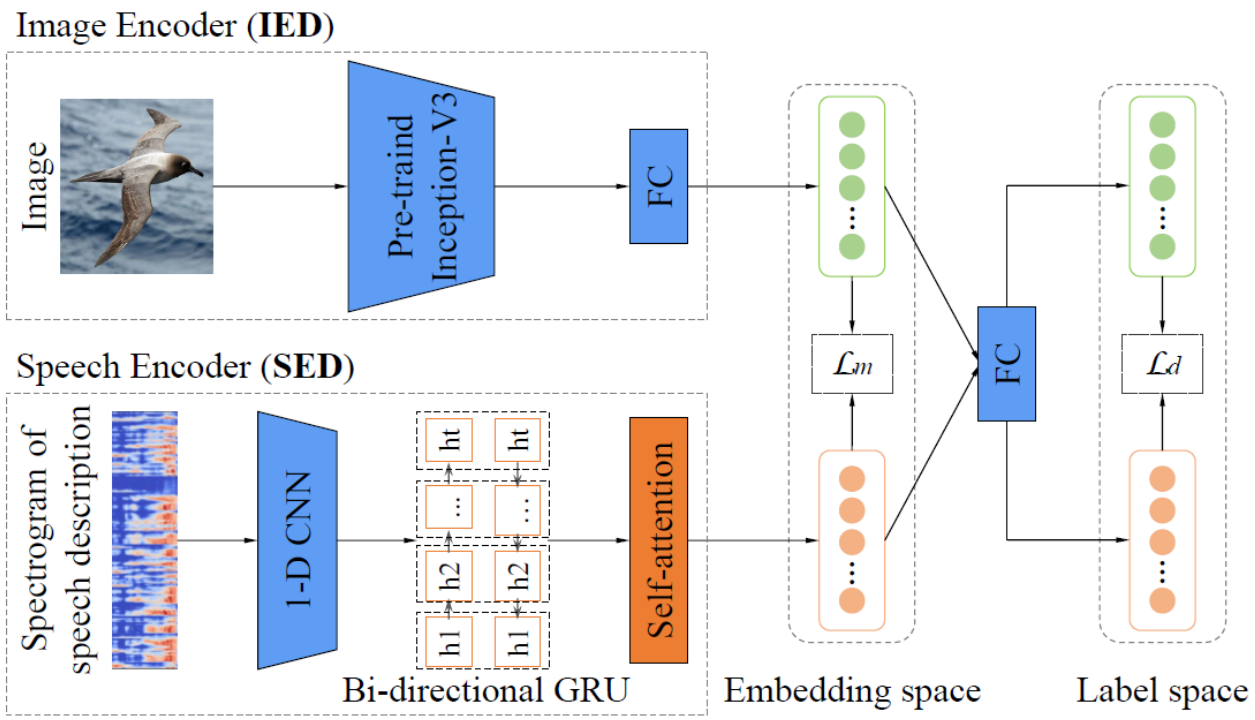
Two stages of speech-to-image generation:



Method

Speech embedding

The speech encoder is trained in a visually grounded way.



Framework of the speech embedding network (SEN).

- Image encoder (IED):

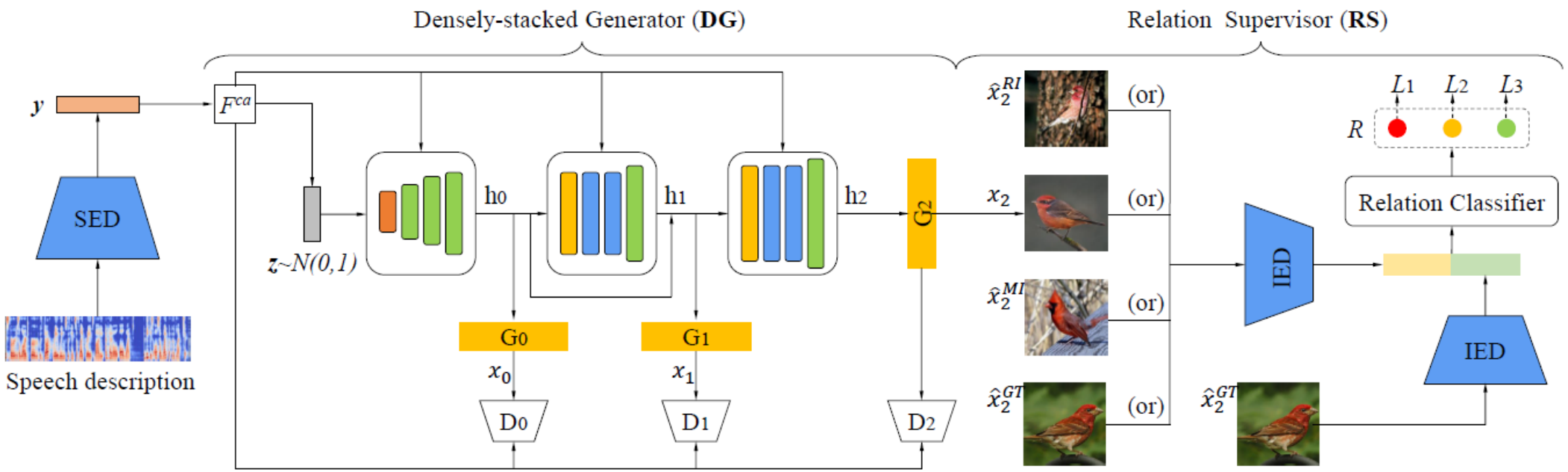
  To obtain image embeddings

- Speech encoder (SED):

  To obtain speech embeddings

Method

Generation model



- Densely-stacked generator

  Generates images from low-resolution to high-resolution step by step.

- Relation Supervisor

  To ensure that the generator produces high-quality images that are semantically aligned with the spoken description.

Experiments

Database

CUB-200 Bird

- " A bird with a blue head and a red breast with black wingbars. "
- " This bird has a bright blue crown and a brownish/red breast color. "
- …

Statistics	Training set	Test set	Total
class	150	50	200
image	8855	2933	11788

Oxford-102 Flower

- "This flower has small pink petals with a yellow center. "
- "this flower has small pink petals with a yellow center. "
- …

Statistics	Training set	Test set	Total
class	82	20	102
image	7034	1155	8189

Speech Caption:

Each image of CUB and Oxford has 10 text captions but no speech caption.

Speech captions are synthesized with Tacotron2 with text captions as input.

Experiments

Evaluation

- Inception Score (IS, higher is better)
  - ✓ higher IS means the model can generate more diverse and meaningful images
- Fréchet inception distance (FID, lower is better)
  - ✓ lower FID means a smaller distance between the generated and real image distributions
- mean Average Precision (mAP, higher is better)
  - ✓ higher mAP means higher semantic consistency between generated images and their corresponding speech descriptions.

Method

Results

Evaluation Metric	Input	CUB (Bird) mAP	CUB (Bird) FID	CUB (Bird) IS	Oxford-102 (Flower) mAP	Oxford-102 (Flower) FID	Oxford-102 (Flower) IS
StackGAN-v2	text	—	15.30	4.04±0.05	—	48.68	3.26±0.01
StackGAN-v2†	text	7.01	20.94	4.02±0.03	9.88	50.38	3.35±0.07
MirrorGAN	text	—	—	4.56±0.05	—	—	–
SEGAN	text	—	—	**4.67±0.04**	—	—	–
[12]	speech	—	18.37	4.09±0.04	—	54.76	3.23±0.05
StackGAN-v2	speech	8.09	18.94	4.14±0.04	12.18	54.33	**3.69±0.08**
S2IGAN	speech	**9.04**	**14.50**	4.29±0.04	**13.40**	**48.64**	3.55±0.04

StackGAN-v2: zhang et al, 2018 IEEE TPAMI.

MirrorGAN: qiao et al, 2019 CVPR.

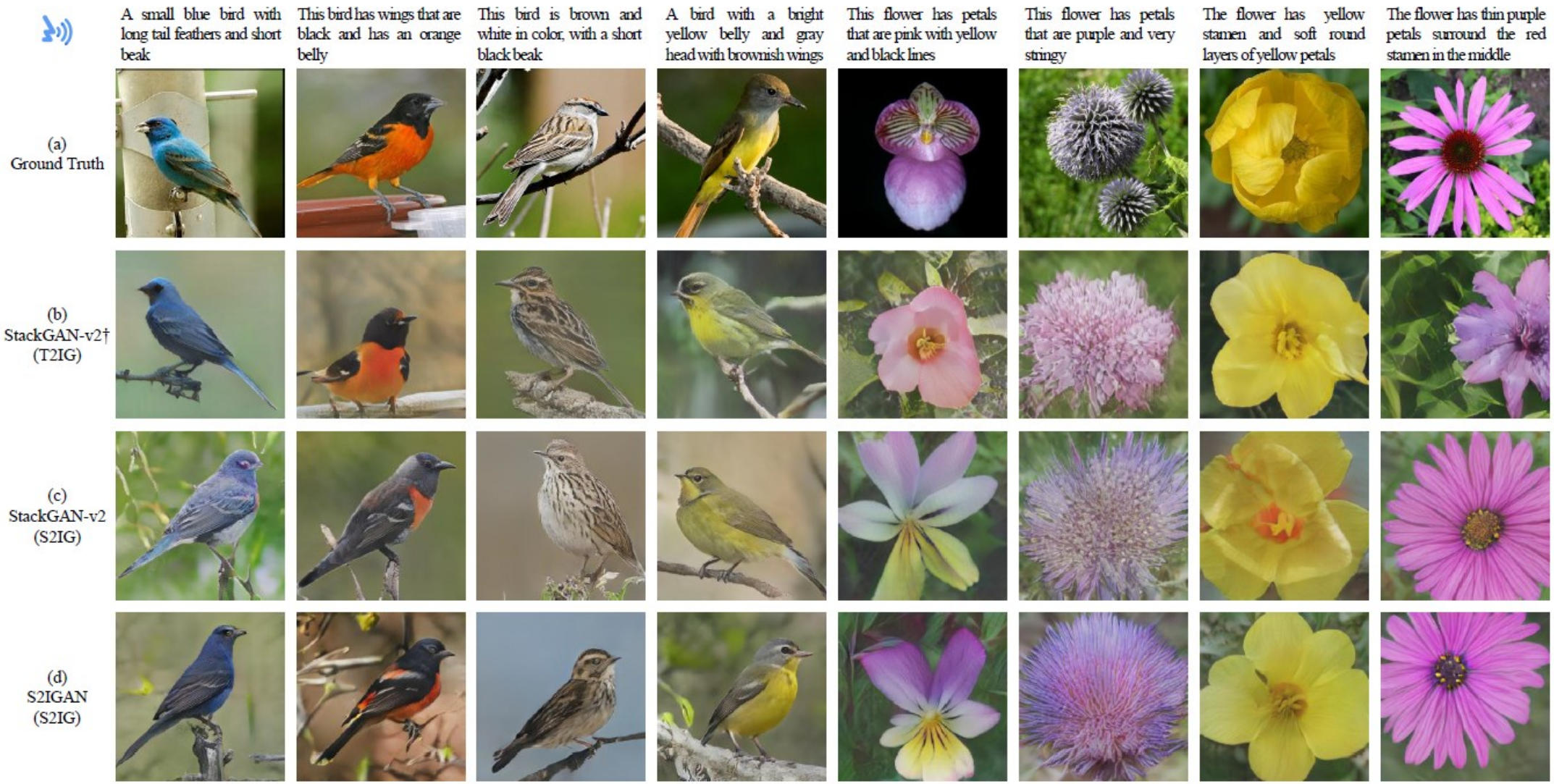
SEGAN: tan et al, 2019 ICCV.

[12]: li et al, "Direct speech-to-image translation", 2020 arXiv

- On the speech-to-image task, our method achieves the state-of-the-art performance.
  - ✓ establishes a solid new baseline for the S2IG task
- StackGAN-v2 that takes our speech embeddings as input outperforms the original text-based image generation.
  - ✓ the learned speech embeddings are competitive compared to text embeddings
  - ✓ the proposed speech embedding model is effectiveness

Method

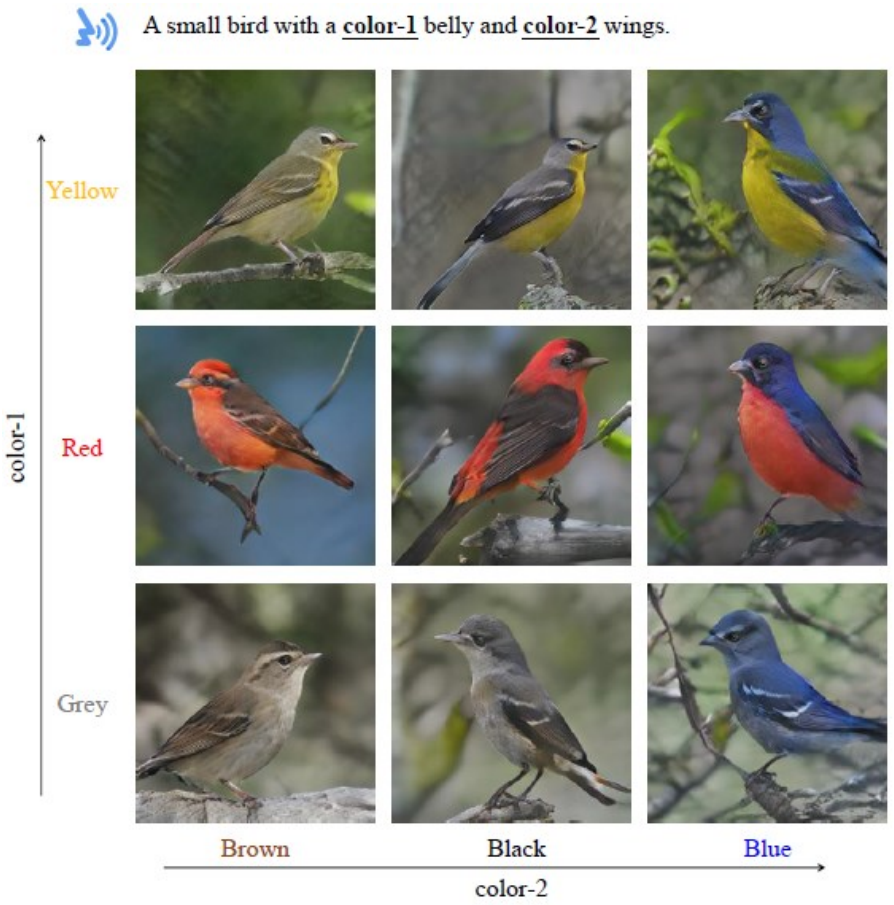
Results



Method

Results

S2IGAN's ability to catch subtle semantic differences in the speech descriptions



- Generated images conditioned on different speech descriptions in which color keywords were changed, e.g.,
  - ✓ A small bird with a **Yellow** belly and **Brown** wings.
  - ✓ A small bird with a **Yellow** belly and **Black** wings
  - ✓ …

Project page

Conclusion and future work

Conclusion

- A novel speech-to-image generation (S2IG) task is proposed.
- The proposed speech-to-image generation model (S2IGAN) is able to generate high-quality images.
- S2IGAN achieves the state-of-the-art performance on the speech-to-image task.

Future work

- Use natural speech instead of synthesized speech database.
- Develop automatic speech word unit discovery method to make the speech-to-image generation system can benefit from the word-level attention mechanism.

Thanks for your attention!